

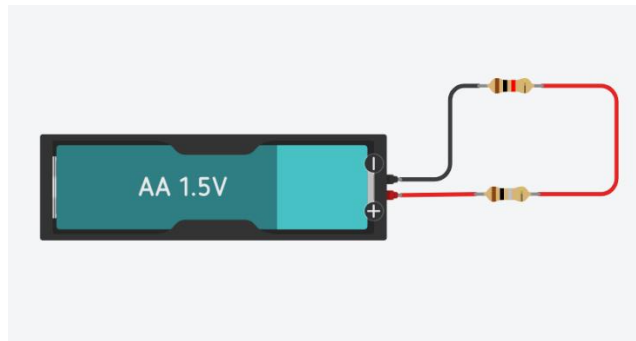
Internal Resistance

ENGS124 Electricity and Magnetism Lab

25 November 2025

Theory

All batteries in the real world have internal resistances. They can be modeled by a series of resistor. If we connect an external resistor to this battery, we have the following schematic:



After applying Kirchoff's voltage law, the following equation is reached:

$$\varepsilon = I(r_i + R) \Rightarrow V_R = \varepsilon - r_i I$$

The experiment was carried out to test the equation and find the internal resistance of the battery.

Experiment

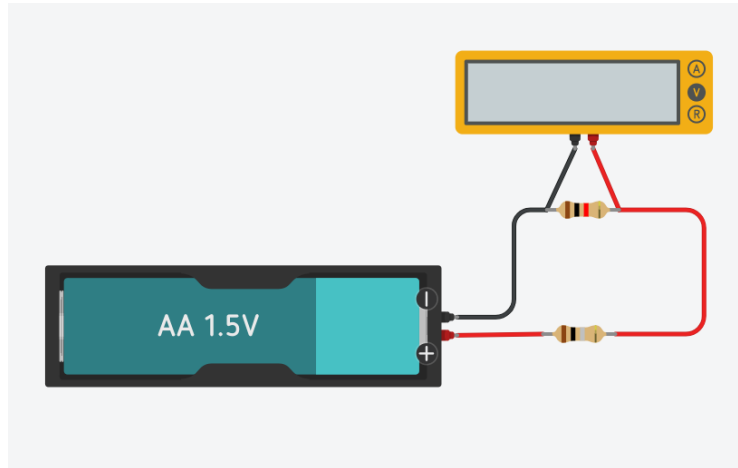
The following circuit was assembled, with li-ion battery and a set of resistors. The voltage across the resistors were measured, as well as the voltage across the battery, before connecting the resistors. Voltage of the battery was measured to be 3.86 V, hence resistors with 5 W working power were chosen using simple calculations.

$$P = \frac{V^2}{R}$$

$$R = \frac{V^2}{P} = \frac{(3.86)^2}{5} = 2.98 \, \Omega$$

Any resistor with greater than 2.98 Ω will run under 5 W and be safe.

Setup



Instruments

Name	Quantity	Properties
Multimeter	1	Voltage and resistance measuring device
Resistors	~	Set of resistors
Connection Wires	~	Copper wires
Breadboard	1	A platform for assembling the circuit
Battery and holder	1	A charged li-ion battery

Variables

Name	Type	Resolution
R	Control	0.1Ω
V_R	Measure	0.01 V
V_{Battery}	Parameter	0.1 V

Errors

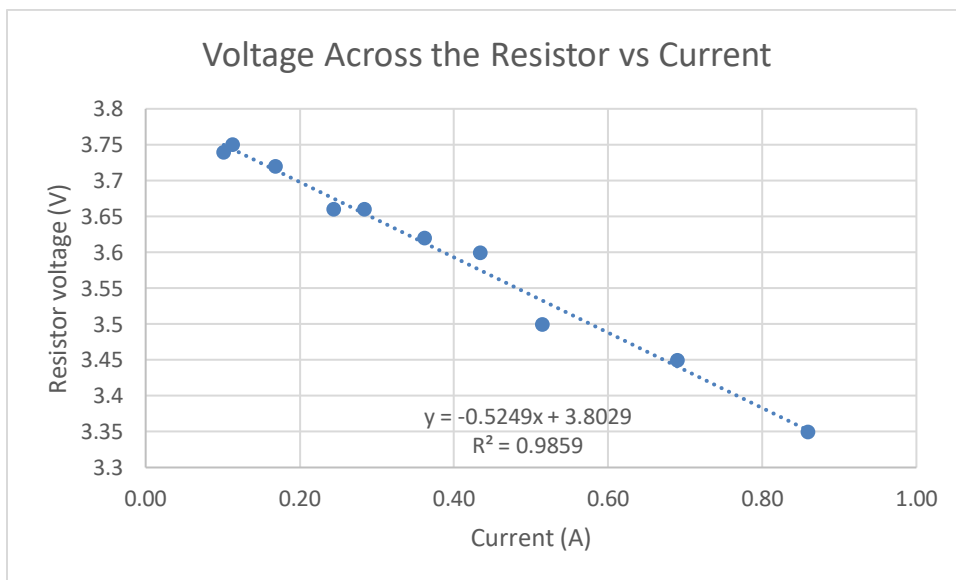
Name	Type	Impact
Voltmeter Internal Resistance	Systematic	Voltmeter itself has some internal resistance and while connected in parallel to the resistor it might affect the readings.

Cable Resistance	Systematic	Since we are using low value resistors, the cable resistances can distort the results
Resistor Heating	Systematic	The resistor's resistance changes when it heats up.

Data

$V_{Battery}$	$R \ \Omega$	V_R	I
3.86	3.9	3.35	0.86
3.86	5	3.45	0.69
3.86	6.8	3.5	0.51
3.86	8.3	3.6	0.43
3.86	10	3.62	0.36
3.86	12.9	3.66	0.28
3.86	15	3.66	0.24
3.86	22.1	3.72	0.17
3.86	33.2	3.75	0.11
3.86	36.9	3.74	0.10

Plot



As we can see above, the relationship between resistor voltage and current is confirmed and the theoretical formula is valid.

The value of E given by the graph is 3.8 V which has 1.55% error compared to the measured value. Internal resistance corresponds to the slope of the graph which is $r_i = 0.52 \, \Omega$.